

Computer Science & IT

Discrete Mathematics

Mathematical Logic

Lecture No. 03

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Recap of Previous Lecture



✓ **Topic**

Logical Implication

✓ **Topic**

Logical Equivalences

✓ **Topic**

Argument / Inference

✓ **Topic**

Rules of inference

✓ **Topic**

Practice questions

Topics to be Covered



✓ Topic

Practice questions

✓ Topic

Proof by contradiction

✓ Topic

Conditional proof

Today's Topic

H.W Q. Simplify the following Statement

$$\{ \sim P \wedge (\sim q \wedge r) \} \vee \{ (q \wedge r) \vee (P \wedge r) \}$$

$$\{ (\sim P \wedge \sim q) \wedge r \} \vee \{ (q \vee P) \wedge r \}$$

$$\{ \underbrace{\sim(P \vee q)} \wedge r \} \vee \{ \underbrace{(P \vee q)} \wedge \underline{\sim r} \}$$

$$\{ \{ \sim(P \vee q) \vee (P \vee q) \} \wedge r \}$$

$$\{ T \wedge r \}$$

$$\equiv r$$

- (a) True
- (b) False
- (c) p
- (d) q
- ☒ (e) r

H.W. Q: Simplify the following Expression

$$[(P \vee Q) \wedge \sim \{ \sim P \wedge (\sim Q \vee \sim R) \}] \vee [(\sim P \wedge \sim Q) \vee (\sim P \wedge \sim R)]$$

$$[(P \vee Q) \wedge \{ P \vee (Q \wedge R) \}] \vee [\sim P \wedge (\sim Q \vee \sim R)]$$

$$[(P \vee Q) \wedge \{ (P \vee Q) \wedge (P \vee R) \}] \vee [\sim P \wedge (\sim Q \vee \sim R)]$$

$$[(P \vee Q) \wedge (P \vee R)] \vee \sim [P \vee (Q \wedge R)]$$

$$\underbrace{[P \vee (Q \wedge R)]}_X \vee \sim \underbrace{[P \vee (Q \wedge R)]}_X$$

$\equiv \text{True}$

H.W. Q: Which of the following is/are valid?

✓ (a) $(\overset{T}{P} \wedge \overset{T}{Q})$ logically implies $(\overset{T}{P} \leftrightarrow \overset{T}{Q}) \equiv \overset{T}{T}$

$\checkmark$ (b) $(P \leftrightarrow Q)$ logically implies $(P \rightarrow Q)$

~~⊗~~ $(P_F \leftrightarrow Q_F)$ logically implies $(P_T \rightarrow F \equiv F)$
 $F \rightarrow T \equiv T$

~~⊥~~ $\mathcal{Q}$ logically implies $(P \rightarrow \mathcal{Q})$
 $\emptyset \rightarrow T = T$

May be false
∴ Invalid

H.W. Q: Which of the following is/are tautology?

☒ a $(P \wedge Q) \longrightarrow (P \leftrightarrow Q)$

☒ b $(P \leftrightarrow Q) \longrightarrow (P \longrightarrow Q)$

☐ c $(P \leftrightarrow Q) \longrightarrow (P \longrightarrow \sim Q)$

☒ d $Q \longrightarrow (P \longrightarrow Q)$

H.W. Q: Which of the following is/are tautology?

X (a) $(a \vee b) \rightarrow (b \wedge c)$
 $T \vee F = T \quad F \wedge \emptyset = F$

✓ (b) $(a \wedge b) \rightarrow (b \vee c)$
 $T \wedge T \equiv T \quad T \vee \emptyset \equiv T$

X (c) $(a \vee b) \rightarrow (b \rightarrow c)$
 $\emptyset \vee T \equiv T \quad T \rightarrow F \equiv F$

X (d) $(a \rightarrow b) \rightarrow (b \rightarrow c)$
 $\emptyset \rightarrow T \equiv T \quad T \rightarrow F \equiv F$

H.W. Q. Consider the following statements

$S_1: \{(A \wedge B) \rightarrow C\} \equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$

$S_2: \{(A \vee B) \rightarrow C\} \equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$

then which of the following is true?

- (a) Both S_1 & S_2 are valid
- (b) Only S_1 is valid
- (c) Only S_2 is valid
- (d) Neither S_1 nor S_2 is valid.

$$P \equiv Q \quad \text{iff} \quad \begin{array}{l} P \text{ logically implies } Q \\ \uparrow \\ Q \text{ logically implies } P \end{array}$$

H.W. Q. Consider the following statements

$$S_1: \{ \overset{\text{L.H.S.}}{(A \wedge B) \rightarrow C} \} \not\equiv \{ \overset{\text{R.H.S.}}{(A \rightarrow C) \wedge (B \rightarrow C)} \}$$

$$S_2: \{ (A \vee B) \rightarrow C \} \equiv \{ (A \rightarrow C) \vee (B \rightarrow C) \}$$

$$\rightarrow \text{L.H.S.} = \{ (A \wedge B) \rightarrow C \}$$

$$\{ \sim (A \wedge B) \vee C \}$$

$$\{ (\sim A \vee \sim B) \vee C \} = \{ (\sim A \vee C) \vee (\sim B \vee C) \}$$

$$\equiv C \vee C$$

$$\equiv \{ (A \rightarrow C) \vee (B \rightarrow C) \} \neq \text{R.H.S.}$$

H.W. Q. Consider the following statements

$$S_1: \{ (A \wedge B) \rightarrow C \} \equiv \{ (A \rightarrow C) \wedge (B \rightarrow C) \}$$

$$S_2: \{ (A \vee B) \rightarrow C \} \equiv \{ (A \rightarrow C) \vee (B \rightarrow C) \}$$

$$\text{L.H.S.} = \{ (A \vee B) \rightarrow C \}$$

$$= \{ \sim (A \vee B) \vee C \}$$

$$= \{ (\sim A \wedge \sim B) \vee C \} = \{ (\sim A \vee C) \wedge (\sim B \vee C) \}$$

$$= \{ (A \rightarrow C) \wedge (B \rightarrow C) \} \neq \text{R.H.S.}$$

H.W. 8.

then which of the following is true ?

- (a) Both S_1 & S_2 are valid
- (b) Only S_1 is valid
- (c) Only S_2 is valid
- ~~(d) Neither S_1 nor S_2 is valid.~~

$$P \equiv Q \quad \text{iff} \quad \begin{array}{l} P \text{ logically implies } Q \\ \uparrow \\ Q \text{ logically implies } P \end{array}$$

H.W. Q. Which of the following is/are true?

✓ a) $\{a \rightarrow (b \vee c)\} \equiv \{(a \wedge \sim b) \rightarrow c\}$

$$a \rightarrow (b \vee c) = \sim a \vee (b \vee c) = (\sim a \vee b) \vee c = \sim(a \wedge \sim b) \vee c = (a \wedge \sim b) \rightarrow c$$

= RHS

✓ b) $\{(P \rightarrow Q) \wedge (R \rightarrow Q)\} \equiv \{(P \vee R) \rightarrow Q\}$

{ from proof of S₂ }
in previous question }

✓ c) $\{(P \rightarrow Q) \vee (R \rightarrow Q)\} \equiv \{(P \wedge R) \rightarrow Q\}$

{ from proof of S₁ }
in previous question }

✓ d) All of the above

H.W. Q. Consider the following propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

Above propositional statement is.

- (a) Satisfiable but not valid {i.e., Contingency}
- (b) Valid
- (c) Contradiction

H.W.

Q. Consider the following propositional statement.

$$\{ P \rightarrow (Q \vee R) \} \longrightarrow \{ (P \wedge Q) \rightarrow R \}$$

in that case

$$P \rightarrow (Q \vee R)$$
$$T \rightarrow (T \vee F) = T$$

L.H.S is true

$\therefore$ for $P=T, Q=T \& R=F$

Expression can be false

As well

$\therefore$ Not valid

$\hookrightarrow$ if $R = \text{True}$ then R.H.S. is true and hence.

Whole implication will be true
 $\therefore$ True for at least one case

$\hookrightarrow$ if $P=T, Q=T \& R=F$
then only R.H.S. can be false

H.W.

Q. Consider the following propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

Above propositional statement is.

- ☒ (a) Satisfiable but not valid {i.e., Contingency}
- ☐ (b) Valid
- ☐ (c) Contradiction

H.W. Q:- Check whether the following argument is valid or invalid

* If today is Vishal's B'day, then today is 1st April

* Today is 1st April

∴ Today is Vishal's B'day

let P = Today is Vishal's B'day

Q = Today is 1st April

∴ Argument becomes =

$P \rightarrow Q$

Q

invalid

∴ P

Fallacy of affirming the consequent

H.W. Q:- Check whether the following argument is valid or invalid

- * If today is Vishal's B'day, then today is 1st April
- * Today is not 1st April

∴ Today is not Vishal's B'day

p = Today is Vishal's B'day

$\sim p$ = Today is not Vishal's B'day

q = Today is 1st April

$\sim q$ = Today is not 1st April

Argument is

$p \rightarrow q$

$\sim q$

$\therefore \sim p$

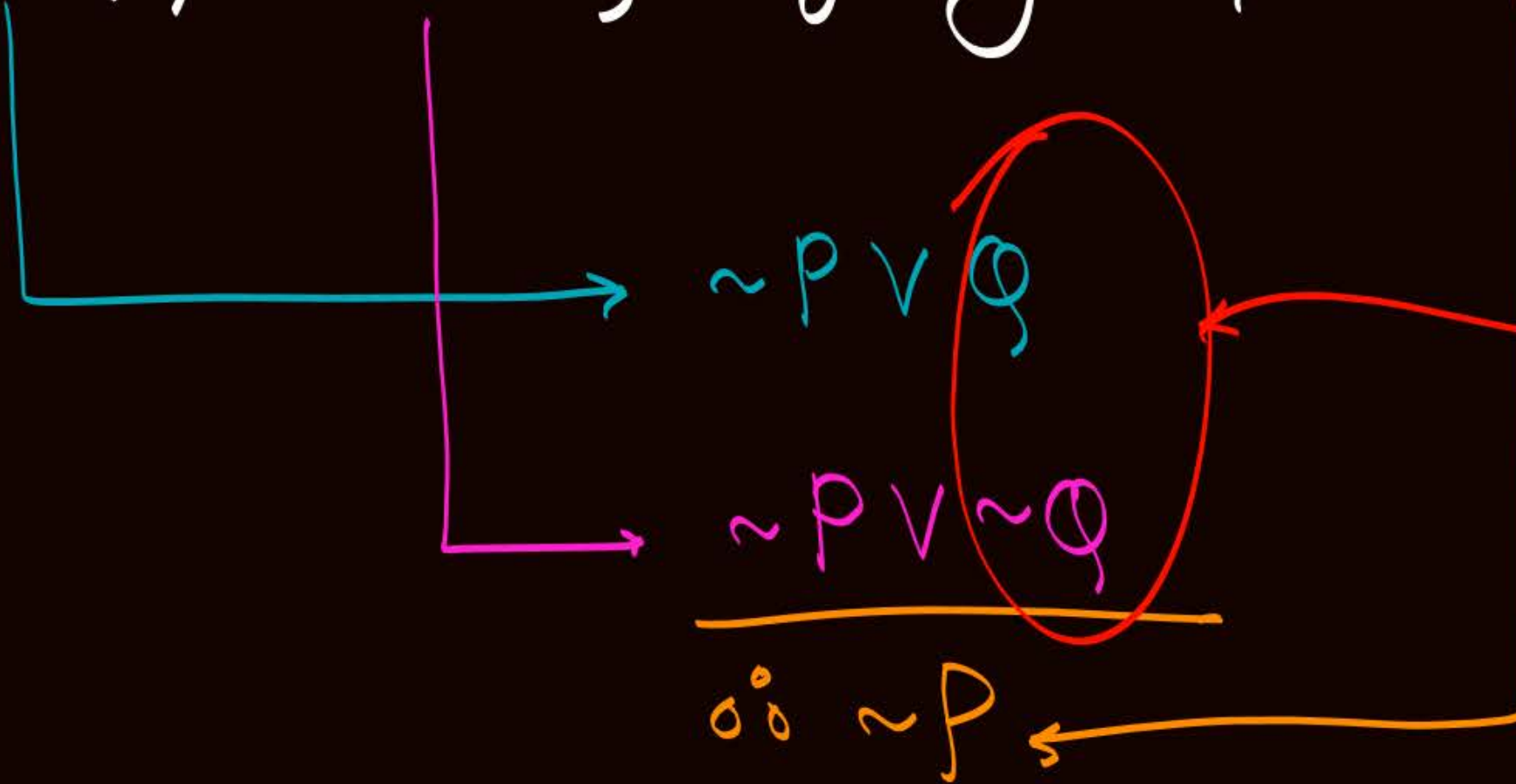
is valid { Modus
Tollen's }

H.W.
★

Q:.

Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ logically implies $\sim P$



Resolution Rule

At least one of
 Q or $\sim Q$
is false
 $\therefore \sim P$ must
be true



Topic : Proof by contradiction

If we want to check whether the argument $\{P_1, P_2, P_3, \dots, P_n\} \vdash Q$ is valid or not

→ then in order to check for the validity of the argument we will assume that the conclusion of the argument is false {i.e. we will assume that Q is false, i.e. we will assume that ' $\sim Q$ ' is true}, and will include this false conclusion in the given set of premises.

∴ Our new set of premises will be $\{P_1, P_2, P_3, \dots, P_n, \sim Q\}$

• If this assumption leads to any contradiction, then our assumption is wrong {we assumed that ' $\sim Q$ ' is true and it is wrong means ' Q ' is true}, and hence given argument is valid, if no contradiction, then invalid argument

Q: Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ logically implies $\sim P$

Solution using proof by contradiction

Conclusion of the argument is ' $\sim P$ '

① Assume that Conclusion of the argument is false
i.e., assume that ' $\sim P$ ' is false, i.e., assume that ' P ' is true

② Include this false Conclusion in the set of premises

∴ New set of premises will be

$\{P \rightarrow Q, \sim(P \wedge Q), P\}$

∴ Q (by M.P.)

$\sim P \vee \sim Q$ (by De Morgan's Law)

D.S.

∴ $\sim P$

it is contradiction
∴ Our assumption is wrong
Hence, Argument is valid.

Q:- Check whether the following argument is valid or not?

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q\}$ logically implies " $R \rightarrow S$ "

① Assume Conclusion is false, i.e. assume $(R \rightarrow S)$ is false,

② i.e. assume $\sim(R \rightarrow S)$ is true

∴ Our new set of premises will be

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q, \sim(R \rightarrow S)\}$

$\equiv$
 $\sim(\sim R \vee S)$

$\equiv$
 $R \wedge \sim S$

$\equiv$
 $R, \sim S$

$Q \rightarrow S$

DS.

P

M.P.

∴ S

Contradiction
Hence Argument is valid

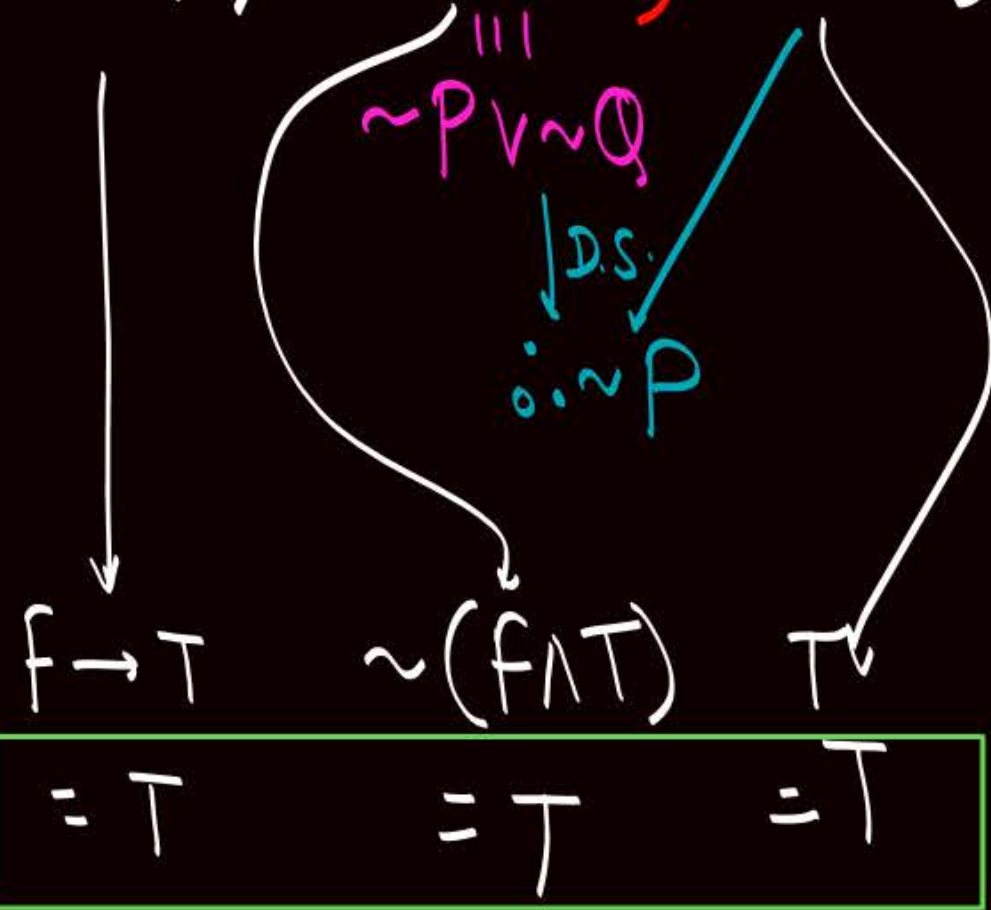
Q: Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ logically implies $\sim Q$

① Assume Conclusion is false, i.e., $\sim Q$ is false, i.e., Q is true

② $\therefore$ New set of premises will be

$\{P \rightarrow Q, \sim(P \wedge Q), Q\}$



When $P = F, \& Q = T$;
then all premises
can be true

All premises can be true
i.e. No Contradiction
Hence, Argument is invalid



Topic : Conditional proof

The argument
 $\{P_1, P_2, P_3, \dots, P_n\} \vdash Q \rightarrow R$ is valid
if and only if, the argument
 $\{P_1, P_2, P_3, \dots, P_n, Q\} \vdash R$ is valid

i.e., checking whether the argument
 $\{P_1, P_2, P_3, \dots, P_n\} \vdash Q \rightarrow R$ is valid or not
is exactly same as checking
whether the argument
 $\{P_1, P_2, P_3, \dots, P_n, Q\} \vdash R$ is valid or not

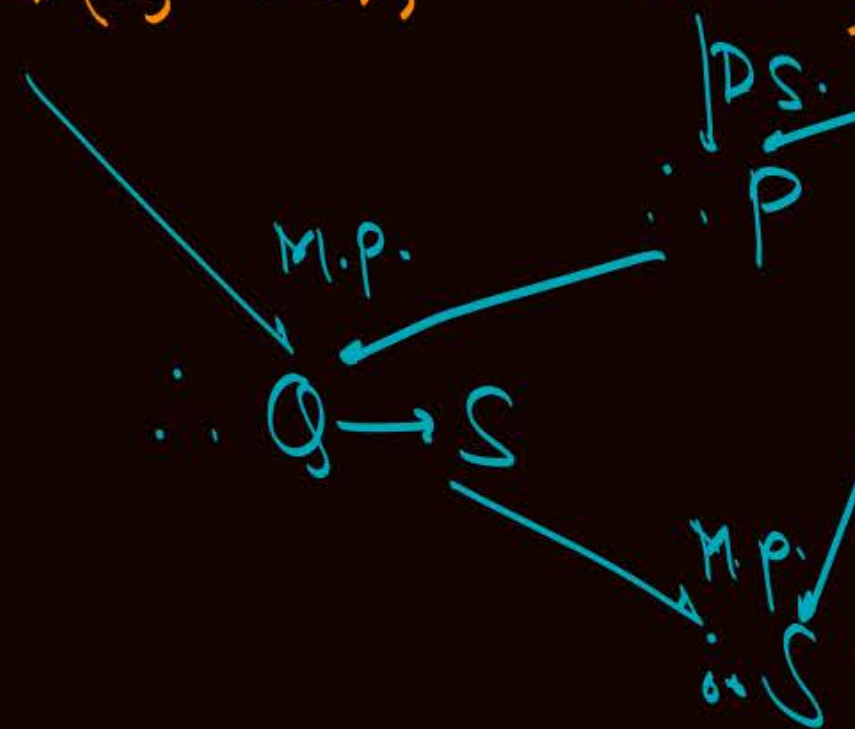
Q:- Check whether the following argument is Valid or not?

Soln using Conditional proof
 $\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q\}$ logically implies " $R \rightarrow S$ "

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q\}$ logically implies " $R \rightarrow S$ "

if and only if

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q, R\}$ logically implies S



Hence given argument is Valid

Q:- Check whether the following argument is
Valid or invalid

If A works hard, then B or C will enjoy themselves

If B enjoys himself, then A will not work hard

If C enjoy himself, then D will not enjoy himself

∴ If A works hard then D will not enjoy himself.

Q:- Check whether the following argument is Valid or invalid

If A works hard, then B or C will enjoy themselves $\equiv a \rightarrow (b \vee c)$

If B enjoys himself, then A will not work hard $\equiv b \rightarrow \sim a$

If C enjoy himself, then D will not enjoy himself $\equiv c \rightarrow \sim d$

$\therefore$ If A works hard then D will not enjoy himself $\equiv \therefore a \rightarrow \sim d$

Let,
a : 'A' works hard
b : 'B' enjoy himself
c : 'C' enjoy himself
d : 'D' enjoy himself.

Argument is, $\{a \rightarrow (b \vee c), b \rightarrow \sim a, c \rightarrow \sim d\} \vdash a \rightarrow \sim d$
Given argument is Equivalent to the argument
 $\{a \rightarrow (b \vee c), b \rightarrow \sim a, c \rightarrow \sim d, a\} \vdash \sim d$
is valid



Topic : Some important statements

★ 1. $P \rightarrow Q \equiv Q$ follows from P

2. P implies $Q \equiv P \rightarrow Q$

3. If P then $Q \equiv P \rightarrow Q$

4. P only if $Q \equiv P \rightarrow Q$

5. P is sufficient condition for $Q \equiv P \rightarrow Q$

6. Q is necessary condition for $P \equiv P \rightarrow Q$



Topic : Some important statements



→ 7. $P \text{ if } Q \equiv Q \rightarrow P$

8. $P \text{ when } Q \equiv Q \rightarrow P$

9. $P \text{ follows from } Q \equiv Q \rightarrow P$

10. $P \text{ unless } \sim Q \equiv Q \rightarrow P$

$$P \vee \sim Q \equiv Q \rightarrow P \equiv \sim P \rightarrow \sim Q$$

you ^P can not clear gate

unless

you appear for gate

Q ✓

~P

|||

If you cleared gate

then

you must have appeared for gate

Q ✓

P unless Q ✓

|||

~P → Q ✓

|||

P ∨ Q ✓

ie unless is
same as OR(∨)



2 mins Summary



✓
Topic

Practice questions

✓
Topic

Proof by contradiction

✓
Topic

Conditional proof

THANK - YOU